MASTER OF BUSINESS ADMINISTRATION

SEMESTER 1



DMBA117

DATA VISUALIZATION

Unit – 8

Advanced Visualizations Techniques in Tableau

TABLE OF CONTENTS

SL No	Topic	Fig No / Table / Graph	SAQ / Activity	Page No
1	Introduction	-	-	4
	1.1 Objectives	-	-	
2	Calculated Fields and Parameters	-	-	5 - 10
	2.1 Calculated Fields	1	-	
3	Using Parameters	2	-	11 - 15
4	Advanced Charts in Tableau	-	-	16 - 32
	4.1 Bullet Charts	3	-	
	4.2 Heat Maps	4	-	
	4.3 TreeMaps	5	-	
	4.4 Waterfall Charts	6	-	
	4.5 Sankey Diagrams	7	-	
	4.6 Scatter Plot with Trend Lines	8	-	
	4.7 Combining Techniques	-	-	
5	Summary	-	-	33
6	Glossary	-	-	34 - 35
7	Self-Assessment Questions	-	1	36
8	Terminal Questions	-	-	37
9	Answers	-	-	38 - 42
10	Suggested Books and References	-	-	43

1. INTRODUCTION

In modern information analysis, data visualization is essential, offering a graphical avenue for conveying complex data and insights. Through the expert utilization of visual elements such as charts, graphs, and maps, data visualization tools become the channel that transforms raw data into a visually clear narrative. This approach offers a potent means of rendering trends, outliers, and patterns observable, thus empowering users to easily grasp the details of data.

In conclusion, data visualization is not merely an auxiliary component of data analysis; it's a transformative approach that complements the worlds of information and perception. It empowers individuals across a spectrum of technical expertise to derive meaning from complex datasets, accelerating decision-making processes and fostering a more comprehensive understanding of data-driven narratives. In the era of Big Data, where information overload is a pressing concern, data visualization emerges as a vital navigational tool, guiding us through the intricacies of data and transforming them into insights that fuel progress and innovation.

1.1. Objectives

After studying this unit, you should be able to:



- Recall fundamental concepts of data visualization, including terminology, data types and common graphical representations.
- Explain the relationship between data visualisation and effective communication. Compare and contrast different visualization techniques, discussing their strengths and weaknesses in conveying specific types of information.
- Demonstrate the ability to choose appropriate visualization types based on the characteristics of the data and the communication goals.
- Justify the choice of visualization methods by explaining how they enhance understanding of specific insights.
- Critique existing data visualizations regarding their accuracy, clarity, and potential biases.

2. CALCULATED FIELDS AND PARAMETERS

Advanced visualization techniques in Tableau empower users to create dynamic, interactive, and meaningful dashboards. These methods enable deeper insights and allow end-users to engage directly with data.

2.1. Calculated Fields

Calculated fields are a cornerstone of Tableau's data analytics capabilities, enabling users to go beyond the raw data provided in their datasets. These fields allow for creating new data points by applying custom formulas and logic to existing fields, effectively transforming raw information into actionable insights. Calculated fields provide the flexibility to perform these tasks within Tableau's intuitive interface, whether you need to compute profit margins, segment customers, or aggregate data based on complex conditions.

One of the most significant advantages of calculated fields is their versatility. They can handle everything from basic arithmetic operations and text manipulations to complex logical conditions and date calculations. This capability allows analysts to customize visualizations to meet specific business requirements, such as identifying trends, comparing performance metrics, or categorizing data dynamically. By enabling the creation of new metrics and dimensions, calculated fields empower users to uncover insights that might otherwise remain hidden.

Moreover, Tableau makes building and validating calculated fields easy through its user-friendly formula editor. The editor includes features like syntax highlighting, built-in functions, and error-checking to simplify the process of creating calculations. As a result, even users without extensive programming experience can leverage calculated fields to enhance their data visualizations. Whether you are an analyst, data scientist, or business professional, mastering calculated fields can dramatically improve the depth and clarity of your Tableau dashboards.

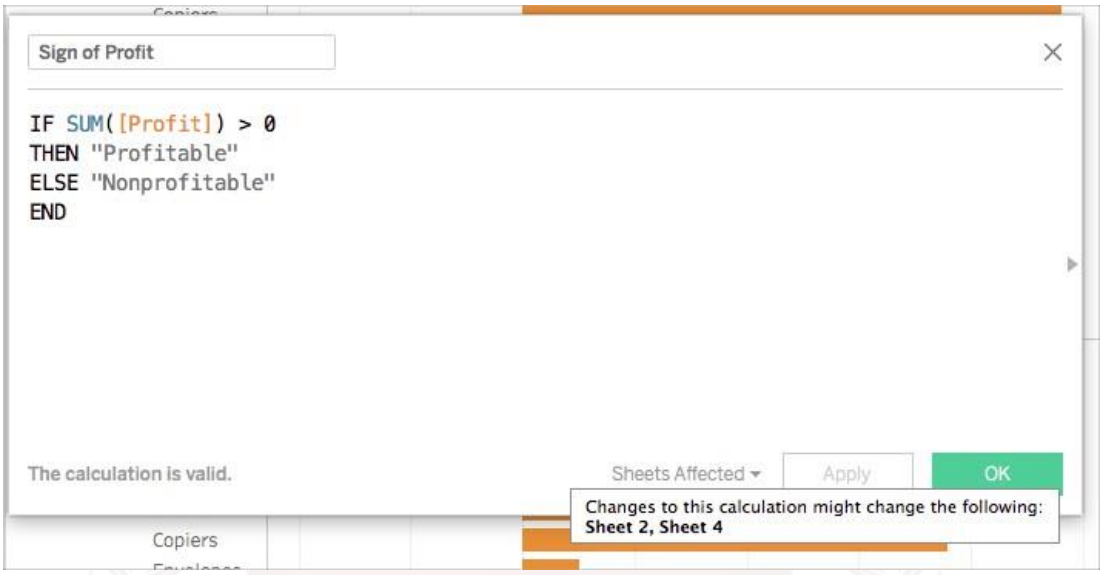


Fig 8.1. Example of the formula editor in Tableau to add Calculated fields

Why Use Calculated Fields?

Calculated fields are one of Tableau's most powerful features. They enable users to go beyond the constraints of raw data to create new metrics, enhance data visualization, and enable dynamic analysis. By transforming, aggregating, or categorizing data, calculated fields allow users to extract deeper insights and customize their dashboards to meet specific business needs.

1. **Data Transformation**
   - One of the primary reasons for using calculated fields is to create new data points or metrics that do not exist in the original dataset. Users can derive additional insights that drive better decision-making by combining or manipulating existing fields.
   - Example: Calculating the profit margin by dividing profit by sales:
     [Profit Margin] = [Profit] / [Sales]
     This derived metric helps businesses evaluate profitability more effectively. For instance, while sales might be high in one region, a low-profit margin could reveal inefficiencies or higher costs, prompting further investigation.
   - Data transformation using calculated fields is also helpful in preparing datasets for advanced analytics or cleaning up messy data. For example, users can split strings, combine fields, or convert data types to align with their analysis needs.

2. **Advanced Aggregation**
   - Calculated fields allow users to perform complex aggregations and summarizations, often combining multiple fields or applying logic beyond basic aggregation options. This

capability is essential for comparing data across different dimensions or summarizing it in ways not available in the raw dataset.

- Example: Displaying average sales for different product categories:
  AVG([Sales])
  Using calculated fields, users can create dynamic summaries that adapt to filtering or user interaction, offering more flexible analysis. For instance, instead of merely showing total sales by category, a dashboard can display the average sales per transaction, offering a deeper understanding of customer behavior.
- Advanced aggregation also supports creating weighted averages, running totals, or moving averages, enabling users to uncover trends and patterns hidden in their data.

3. **Conditional Logic**
   - Conditional logic within calculated fields allows users to dynamically categorize, filter, or highlight data based on specific criteria. This feature is particularly valuable for segmenting data, creating thresholds, or generating insights tailored to specific business rules.
   - Example: Labeling customers as "High Value" or "Low Value" based on their total purchases:
     IF [Total Purchases] > 5000 THEN "High Value"
     ELSE "Low Value"
     END
     With this logic, businesses can easily identify their top customers and develop targeted strategies to retain them. Similarly, Tableau dashboards can highlight underperforming areas, such as products with low sales or regions with minimal profit margins.
   - Conditional logic is also useful for creating alerts or visual indicators. For example, a dashboard could dynamically flag metrics that fall below a target threshold, enabling users to focus on areas that require attention.
4. **Customization**
   - Calculated fields empower users to tailor their dashboards to meet specific needs by modifying the presentation or structure of the data. This customization can involve reformatting fields, performing time-based calculations, or creating new data dimensions.
   - Example: Converting date formats or calculating the number of days between two dates:
     DATEDIFF ('day', [Order Date], [Ship Date])

This calculation helps businesses measure lead times and identify bottlenecks in their processes. Similarly, converting dates into fiscal quarters or calculating year-over-year growth allows businesses to align Tableau dashboards with their operational and reporting standards.

- Customization also extends to user experience. For instance, users can create calculated fields to display shortened labels, merge data for better readability, or adjust numerical formats to suit specific preferences.

Types of Calculations in Tableau

1. Basic Arithmetic:
   - Perform operations like addition, subtraction, multiplication, or division.
   - Example: SUM([Sales]) - SUM([Discount])
2. String Calculations:
   - Manipulate text fields.
   - Example: Concatenating first and last names:
     ([First Name] + " " + [Last Name])
3. Logical Calculations:
   - Implement conditional logic with IF, CASE, AND, OR statements.
   - Example: Categorizing sales performance:
     IF [Sales] > 50000 THEN "High" ELSE "Low" END
4. Date Calculations:
   - Perform operations on date fields.
   - Example: Calculating the difference in days between two dates:
     DATEDIFF('day', [Order Date], [Ship Date])
5. Aggregations:
   - Summarize data at different levels.
   - Example: Calculating total profit:
     SUM([Profit])
6. Table Calculations:

   These are used for advanced data manipulation, such as running totals, ranks, or moving averages.

How to Create a Calculated Field

1. Open the Calculation Editor:

   In the Data Pane, right-click and select Create Calculated Field.
2. Name the Calculated Field:

   Provide a meaningful name that reflects its purpose.
3. Enter the Formula:

   Write your calculation using the built-in formula editor. Tableau provides syntax highlighting and a function list to assist you.
4. Validate the Formula:

   Use the Check button to ensure the syntax is correct.
5. Apply the Field:

   Drag the calculated field to rows, columns, or marks to use it in your visualization.

Examples of Calculated Fields

1. **Profit Margin Calculation**

   Let's say your dataset contains fields for Sales and Profit. To calculate the profit margin:

   Formula: (SUM([Profit]) / SUM([Sales])) * 100

   Steps:

   - Create a new calculated field named Profit Margin.
   - Enter the formula above.
   - Drag the Profit Margin to the visualization to display and format it as a percentage.

2. **Customer Categorization**

   To classify customers into tiers based on their total purchase amount:

   Formula:

   IF SUM([Sales]) > 100000 THEN "Gold"

   ELSEIF SUM([Sales]) > 50000 THEN "Silver"

   ELSE "Bronze" END

   Steps:

   - Create a calculated field named Customer Tier.
   - Use the formula to define categories.

- Use this field in a visualization, such as a pie chart, to display the distribution of customer tiers.

Practical Example: Discounted Sales Calculation

Suppose your dataset contains Sales and Discounts, and you want to calculate the total discounted sales.

Formula: SUM([Sales]) * (1 - AVG([Discount]))

Steps:

- Create a calculated field named Discounted Sales.
- Write the formula above.
- Drag Discounted Sales to your visualization, such as a bar chart, to analyze the impact of discounts across product categories or regions.

Best Practices for Using Calculated Fields

- Test the Calculation: Always validate the formula to avoid errors in your visualizations.
- Use Descriptive Names: Provide meaningful names to help others understand the purpose of the field.
- Optimize for Performance: Avoid overly complex calculations that may slow down your dashboard, especially with large datasets.
- Document Your Work: Provide comments or documentation for complicated calculations.

By mastering calculated fields, you can unlock new dimensions of analysis and derive actionable insights from your data.

3. USING PARAMETERS

Parameters in Tableau are versatile tools that enable dynamic interactivity within dashboards and visualizations. Unlike static filters, which limit data views based on predefined criteria, parameters allow users to input values or select from predefined options to modify calculations, visualizations, or data sources. This flexibility empowers end-users to explore data dynamically, making dashboards more adaptable to different scenarios and analytical needs.

At their core, parameters are single-value variables that can be integrated into calculated fields, filters, reference lines, and more. They support various data types, including numbers, strings, dates, and booleans, and provide several input modes, such as free text, drop-down lists, or sliders. By adjusting a parameter's value, users can directly influence how data is displayed or calculated, such as toggling between metrics, setting thresholds, or performing what-if analyses.

The true power of parameters lies in their ability to enhance user experience and interactivity. For instance, with parameters, you can create a single chart that allows users to switch between visualizing sales, profit, and discount values without needing multiple charts. This not only saves space but also creates a seamless analytical flow. Parameters are particularly useful in scenarios where flexibility and adaptability are crucial, making them a key component in creating dynamic and user-centric Tableau dashboards.

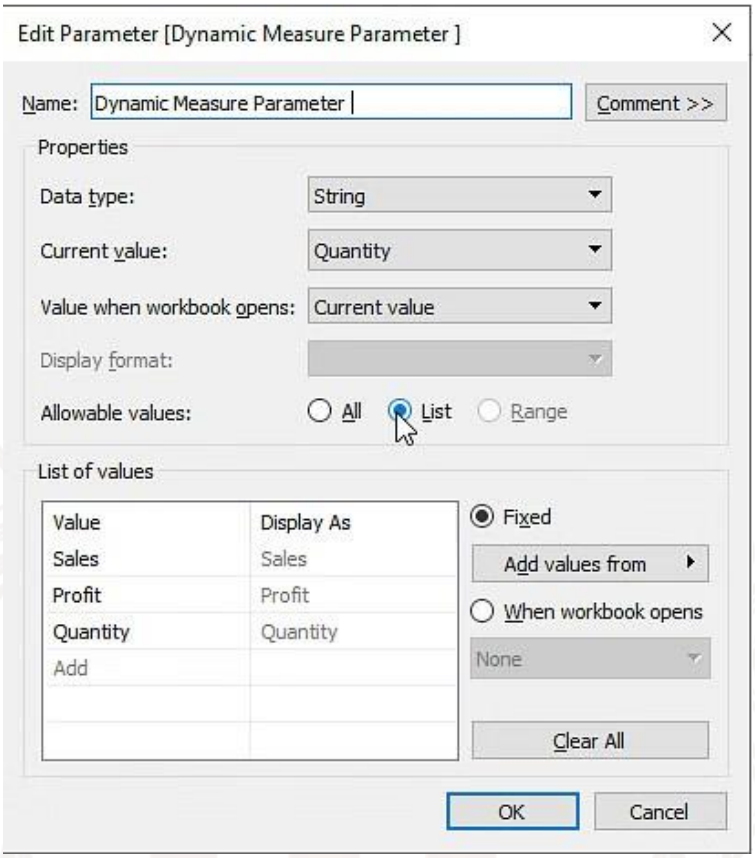


Fig 8.2. Parameter edit menu in Tableau

Key Features of Parameters in Tableau

Parameters in Tableau are a cornerstone of dynamic dashboard design, providing interactivity and customisation to end-users. Here is a detailed look at their key features:

1. **Dynamic Interaction**

   Parameters enhance user interactivity by allowing input values to influence visualizations directly. This capability is particularly valuable for creating adaptable dashboards that cater to diverse analytical scenarios. With parameters, users can adjust thresholds, switch between metrics, or set custom benchmarks in real-time.

   For example, a parameter can let users dynamically define a sales target. This target can then be compared against actual sales, with the visualization updating immediately based on the user-defined input. Similarly, in a scenario involving multiple metrics, a parameter can provide a dropdown menu to toggle between “Sales,” “Profit,” and “Discount,” ensuring the dashboard remains concise and versatile.

   Dynamic interaction through parameters fosters engagement, empowering users to explore data according to their unique requirements without changing the underlying dataset.

2. **Single Value Scope**

   Each parameter in Tableau holds a single value at any given time, making it an effective tool for user-driven selection. Users can input or select this value through a parameter control mechanism: a slider, a text box, or a dropdown menu.

   The single-value nature of parameters makes them precise and predictable, ensuring clarity in their impact on visualizations or calculations. For instance, in a what-if analysis, a parameter can allow users to input a percentage increase or decrease in budget, and the dashboard will update to reflect the corresponding scenarios. This controlled and single-value input mechanism ensures that parameters remain straightforward to use and interpret.

3. **Integration with Calculated Fields**

   One of the most powerful aspects of parameters is their seamless integration with calculated fields. When paired with calculated fields, parameters enable the creation of custom logic that dynamically alters data or visualizations.

   For instance, consider a scenario where a user wants to switch between analyzing total sales and profit margins. By creating a parameter to toggle between these metrics and linking it to a calculated field, you can make a single chart display either metric based on the parameter's value. Similarly, a parameter can define thresholds for conditional formatting, such as highlighting regions with sales above a user-specified value. This integration ensures that dashboards remain adaptable and relevant to varying analytical needs.

4. **Flexible Input Types**

   Parameters in Tableau support a wide range of data types, making them versatile across different use cases. These types include integers, floating-point numbers, strings, booleans, and dates. This flexibility ensures that parameters can be used in diverse scenarios, such as numeric thresholds, text-based category selections, or date range customizations.

   Parameters also offer various modes of value input. For instance, users can:

   - Enter free text or numbers directly.
   - Choose from a predefined list of options (e.g., a dropdown menu of product categories).
   - Use a slider to select a value within a specific range with predefined minimum, maximum, and step sizes.

   Customizing input types ensures that parameters remain intuitive and user-friendly while catering to specific analytical requirements.

Creating Parameters in Tableau

Follow these steps to create and use a parameter:

1. Create the Parameter:
   - In the Data Pane, right-click and select Create Parameter.
   - Give the parameter a descriptive name.
   - Choose the Data Type (Integer, Float, String, Boolean, or Date).
2. Specify the Allowable Values:
   - All: Any value can be entered.
   - List: Predefine a set of allowed values.
   - Range: Define a minimum, maximum, and step size for numeric inputs.
3. Use the Parameter in a Calculated Field:
   - Write a formula incorporating the parameter to adjust values or behavior dynamically.
   - Example: Using a parameter to switch between metrics like sales and profit.
4. Display and Control the Parameter:
   - Right-click the parameter in the Data Pane and select Show Parameter Control.
   - A control widget appears, allowing users to interact with the parameter.

Use Cases for Parameters

- Dynamic Metric Selection: Switch between different measures, such as sales, profit, or discounts.
- What-If Analysis: Model scenarios by adjusting input variables like budget increases or forecast changes.
- Threshold Setting: Allow users to set a custom data segmentation or highlighting threshold.
- Custom Sorting: Enable sorting of categories based on user-selected criteria.

Example: Dynamic Metric Selection

Suppose you want to allow users to toggle between viewing "Sales," "Profit," and "Discount" in a single visualization.

Steps to Implement:

1. Create the Parameter:
   - Name: Metric Selector
   - Data Type: String

- Allowable Values (List):
  - Sales
  - Profit
  - Discount

2. Create a Calculated Field:
   - Name: Selected Metric
   - Formula:

     IF [Metric Selector] = "Sales" THEN [Sales]

     ELSEIF [Metric Selector] = "Profit" THEN [Profit]

     ELSE [Discount] END
3. Use the Calculated Field in the Visualization:
   - Drag the Selected Metric to the Rows shelf.
   - Drag a dimension (e.g., Category) to the Columns shelf.
4. Show the Parameter Control:
   - Right-click Metric Selector in the Data Pane and select Show Parameter Control.
   - A dropdown appears, allowing users to switch between "Sales," "Profit," and "Discount."

Result:

The visualization updates dynamically based on the metric selected in the parameter control, providing a seamless and interactive user experience.

Best Practices for Parameters:

- Combine Parameters with Calculated Fields: This unlocks the full potential of parameters by introducing dynamic logic into your dashboards.
- Keep Parameter Choices Limited: Too many options can overwhelm users. Focus on meaningful and relevant values.
- Use Descriptive Labels: Ensure parameter controls are easy to understand by using clear and concise labels.
- Optimize for Performance: Test parameters with large datasets to avoid performance issues, especially when calculations are complex.

Parameters in Tableau are essential for creating dynamic and user-friendly dashboards. By leveraging them effectively, you can provide users with tailored, interactive visualizations that meet various analytical needs.

4. ADVANCED CHARTS IN TABLEAU

Data storytelling is as much about how data is visualized as it is about the insights it conveys. Advanced chart types in Tableau transform raw data into compelling narratives, enabling users to communicate complex ideas with clarity and precision. These chart types go beyond the traditional bar and line charts by introducing innovative layouts and visual cues that uniquely highlight relationships, patterns, and trends. By leveraging advanced visualisations, analysts can craft dashboards that resonate with audiences and drive informed decision-making.

Tableau's advanced charts include options like Sankey diagrams, bullet charts, waterfall charts, and heat maps, each tailored to specific analytical needs. For example, Sankey diagrams intuitively represent flows and relationships between categories, making them ideal for process or user journey analysis. Bullet charts simplify performance tracking by combining actual values with benchmarks in a single view. Heat maps use color gradients to reveal data density or intensity, helping users quickly identify outliers or clusters within large datasets. These advanced visualizations turn static data into interactive, insightful stories.

The versatility of advanced charts also lies in their ability to integrate with Tableau's other features, such as calculated fields and parameters. This combination allows users to build dynamic, interactive visualizations that adapt to specific questions or scenarios. For instance, a heat map can be paired with calculated fields to highlight high-performing regions, while a parameter-driven bullet chart lets users adjust performance benchmarks in real-time. By using advanced chart types, Tableau users can present data more effectively and foster deeper engagement and understanding among stakeholders.

4.1. Bullet Charts

Bullet charts are compact and effective visualizations designed to compare a primary measure against a benchmark or target value. They are especially useful for performance analysis, enabling users to track progress towards goals clearly and concisely. Unlike traditional bar charts, which only display a single metric, bullet charts incorporate a target indicator, making them ideal for business applications such as sales performance, revenue tracking, or operational efficiency.

Developed by Stephen Few, bullet charts are an excellent alternative to gauges and meters, as they convey more information in less space. Their simplicity allows users to focus on the data while providing critical insights about performance gaps, overachievement, or underachievement. This makes them a staple in dashboards that aim to present actionable insights at a glance.

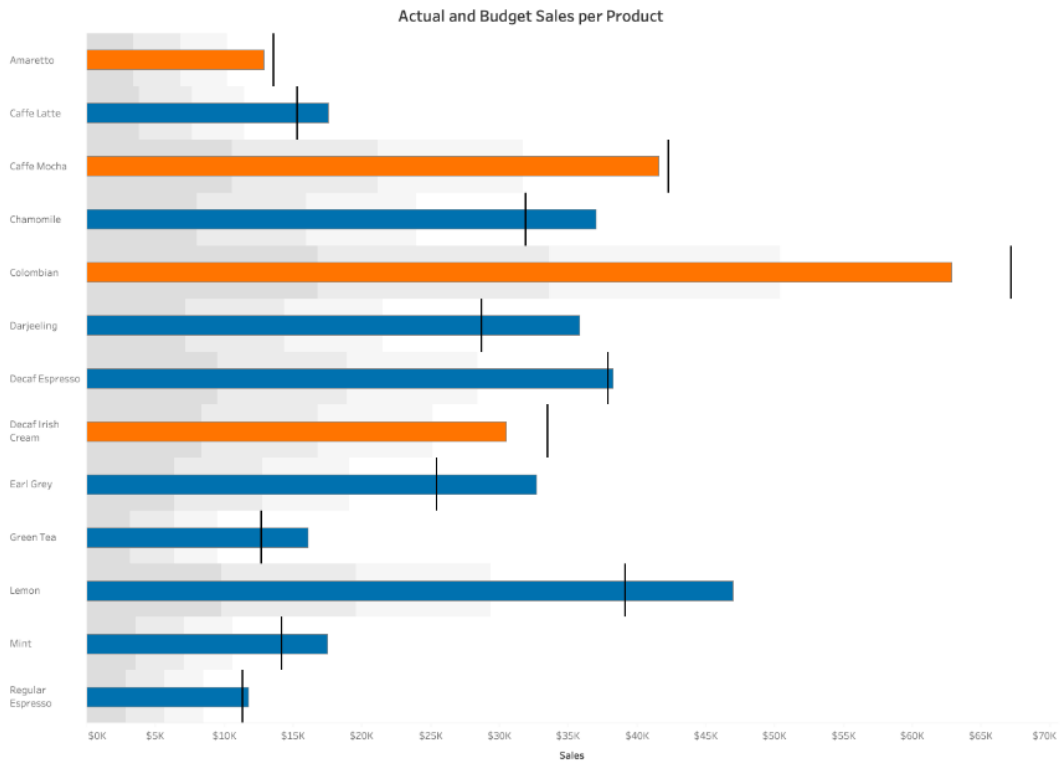


Fig 8.3 *Example of a Bullet chart*

Steps to Create a Bullet Chart in Tableau

Creating a bullet chart in Tableau is straightforward. Follow these steps:

1. **Drag the Primary Measure:**
   Start by dragging a primary measure (e.g., Sales) to the Columns Shelf. This measure represents the actual performance value to be analyzed.
2. **Add a Dimension for Categorization:**
   Drag a dimension (e.g., Region) to the Rows Shelf. This adds granularity to the chart, displaying the primary measure's performance across different categories.
3. **Add the Benchmark or Target Measure:**
   To compare the primary measure against a benchmark, add another measure (e.g., Target Sales) as a reference line. Right-click on the axis of the primary measure, select Add Reference Line and choose the target measure.
4. **Adjust the Marks Card:**
   Set the Marks card to Bar to display the primary measure as a progress bar. Ensure the reference line is visible as a target.

5. **Format the Chart:**

   Adjust the chart's color, size, and layout to customize it. Use contrasting colors to distinguish between the progress bar and the target line for better readability.

Example: Sales Performance by Region

Suppose a retail company wants to evaluate its sales performance across different regions against predefined sales targets. Here's how the bullet chart would look:

1. Data Setup:
   - Primary Measure: Sales (actual performance)
   - Benchmark Measure: Target Sales
   - Dimension: Region
2. Visualization:
   - The chart displays a horizontal bar for each region, representing sales.
   - A vertical reference line within each bar indicates the target sales for that region.
   - For example, if the West region achieves $95,000 in sales against a $100,000 target, the bar will approach the reference line but fall slightly short. Conversely, if the East region surpasses its target, the bar will extend beyond the line.

Insights:

Regions where sales fall short of the target can be quickly identified and analyzed.

Regions exceeding their targets are highlighted, showcasing areas of strong performance.

Bullet charts are an essential visualization tool for comparing performance metrics against benchmarks. By providing a simple yet effective representation of progress, they enable quick identification of performance gaps and areas of success. Their compact design and rich data presentation make them indispensable for dashboards to drive informed decision-making.

4.2. Heat Maps

Heat maps are versatile visualization techniques that use color gradients to represent data density or intensity. They are particularly effective when working with large datasets, as they help identify patterns, trends, or anomalies at a glance. The colors in a heat map range from light to dark or cool to warm, depending on the values being represented. This makes heat maps ideal for comparing multiple variables and highlighting areas of focus in a dataset.

One of the key advantages of heat maps is their ability to transform complex data into an intuitive and

informative visual format. For example, a heat map in business analytics can show sales performance across regions and product categories. Darker or more vibrant colors can signal higher sales, while lighter or muted colors represent lower sales. This visual differentiation allows decision-makers to pinpoint high-performing areas and identify opportunities for improvement quickly.



Fig 8.4 *Example of a Heat Map in Tableau*

Steps to Create a Heat Map in Tableau:

To create a heat map in Tableau, follow these detailed steps:

1. **Set Up Dimensions for Categories:**
   Drag a dimension, such as Category, to the Rows Shelf and another dimension, such as Region, to the Columns Shelf. This creates a grid where each cell represents a unique combination of the dimensions.
2. **Add the Measure for Color Encoding:**
   Drag a measure, such as Sales, to the Color Card on the Marks Shelf. Tableau will automatically apply a color gradient to represent the measure's intensity across the grid.
3. **Adjust the Color Scheme:**
   Use the Color button on the Marks Shelf to customize the gradient. For instance, you can use a warm-to-cool palette (e.g., red for high values, blue for low values) or a sequential gradient (e.g., light to dark shades of a single color). This ensures that the intensity of the data is clearly represented.

4. **Enhance the Visualization:**
    - Drag the same measure (e.g., Sales) to the Text Card to display numeric values within the heat map cells. This provides additional context to the visualization.
    - Optionally, drag a secondary measure (e.g., Profit) to the Size Card to adjust the size of the cells, adding another layer of information.
5. **Format for Clarity:**
    - Adjust the gridlines, borders, and font sizes to ensure readability.
    - Add labels, titles, or annotations as needed to provide context and insights.

Example: Sales Performance by Region and Product Category

A retail chain wants to analyze its sales performance across different regions and product categories. The company's goal is to identify high-performing regions and categories, as well as areas that need improvement.

Data Setup:

- Dimensions: Region (e.g., East, West, North, South) and Category (e.g., Furniture, Technology, Office Supplies).
- Measure: Sales.

Heat Map Visualization:

- Each cell in the grid represents a combination of a region and a category.
- The color intensity corresponds to sales values. For example, a dark red cell in the "Technology" row and "West" column indicates high sales of technology products in the West region.
- Numeric labels within each cell show the exact sales figures, providing precise information alongside the visual representation.

Insights:

- The heat map reveals that "Technology" sales are consistently strong across all regions, with the West region leading.
- "Furniture" sales are weaker in the North and South regions, indicating potential areas for targeted marketing efforts or strategy adjustments.

Heat maps are an invaluable tool for visualizing data density and intensity across multiple dimensions. By leveraging color gradients, they make it easy to identify patterns, trends, and outliers

in complex datasets. Whether used for analyzing sales performance, customer demographics, or operational metrics, heat maps provide actionable insights in a visually compelling format, enabling data-driven decision-making.

4.3. TreeMaps

Tree maps are an innovative way to visualize hierarchical data using nested rectangles. Each rectangle represents a category or sub-category, and its size and color correspond to the values of two measures, such as sales and profit. This makes tree maps highly effective for illustrating part-to-whole relationships, where the contribution of individual elements to the overall total needs to be highlighted. By displaying data hierarchically, tree maps provide a clear visual representation of the distribution of values within a dataset.

Unlike traditional charts, tree maps excel in handling large datasets with multiple levels of categorization. For instance, in business analytics, tree maps can be used to break down sales performance by product categories and sub-categories. The varying sizes and color intensities of the rectangles immediately direct attention to high-performing or underperforming areas, allowing decision-makers to focus on the most impactful segments of their data.

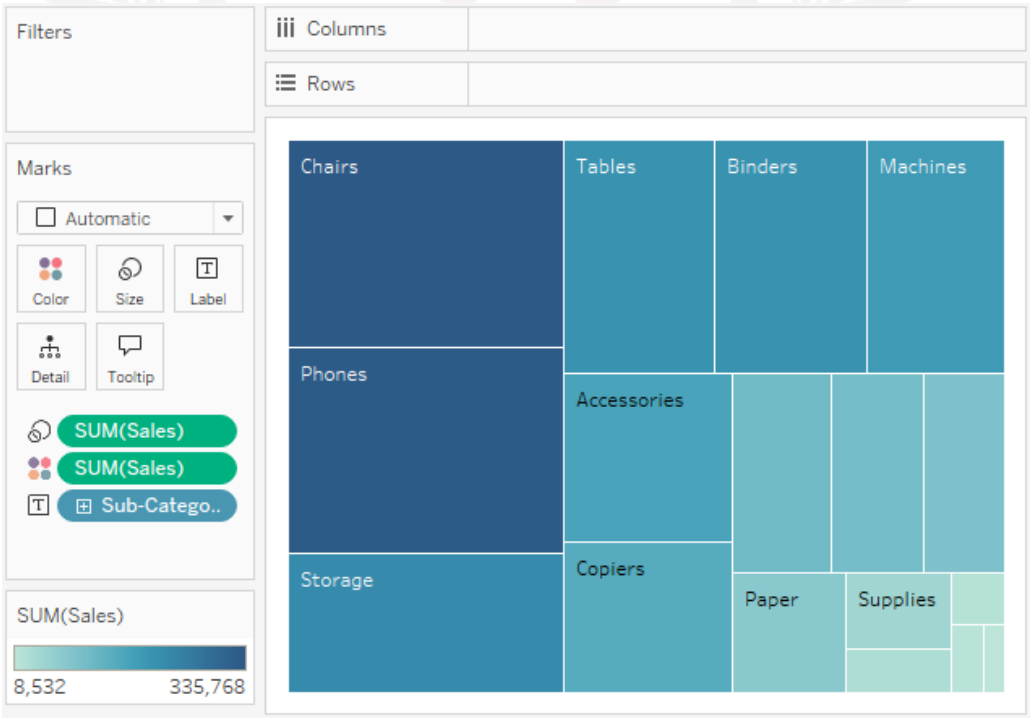


Fig 8.5 *Example of a Tree Map*

Steps to Create a Tree Map in Tableau:

Creating a tree map in Tableau is straightforward. Follow these steps to design an effective

visualization:

1. **Set the Primary Dimension:**

   Drag a high-level dimension, such as Category, to the Rows Shelf. This defines the first layer of the hierarchy.

2. **Add a Sub-Dimension for Detail:**

   Drag a sub-dimension, such as Sub-Category, to the Detail Card on the Marks Shelf. This nests the sub-categories within the primary categories, creating a more granular view of the data.

3. **Assign Measures for Size and Color:**

   - Drag a measure, such as Sales, to the Size Card on the Marks Shelf. This determines the size of each rectangle based on its sales value.
   - Drag another measure, such as Profit, to the Color Card. This applies a color gradient to indicate profitability, with warm or cool tones representing high or low profit margins, respectively.

4. **Customize the Visualization:**

   - Use a diverging or sequential color palette to enhance clarity and highlight variations in profitability.
   - Adjust labels to display key metrics like sales and profit directly within each rectangle for added context.
   - Format the layout to ensure that smaller rectangles remain visible, especially in datasets with large variations in values.

Example: Analyzing Sales and Profit by Product Category

A retail company wants to understand the performance of its products across various categories and sub-categories. The goal is to identify the top contributors to sales and profit, as well as any underperforming areas.

Data Setup:

- Dimensions: Category (e.g., Furniture, Technology, Office Supplies) and Sub-Category (e.g., Chairs, Laptops, Binders).
- Measures: Sales for size and Profit for color.

Tree Map Visualization:

- The largest rectangle might represent the "Technology" category, indicating it has the highest sales. Within this rectangle, smaller rectangles such as "Laptops" and "Phones" represent sub-categories, sized proportionally to their sales values.
- The color intensity of each rectangle reflects profitability. For example, "Laptops" might be a large, dark green rectangle, showing high sales and profit, while "Binders" could be a small, light red rectangle, indicating lower sales and negative profit margins.

Insights:

- The tree map highlights that "Technology" is the most profitable category, with "Laptops" and "Phones" as key contributors.
- In contrast, "Office Supplies" has smaller rectangles with lighter colors, suggesting lower sales and margins. This insight helps the company focus on improving performance in this category.

Tree maps are a powerful tool for visualizing hierarchical data and understanding part-to-whole relationships. By leveraging size and color to represent different metrics, they allow users to quickly identify trends, anomalies, and opportunities within their data. Whether analyzing sales, profit, or any other performance metrics, tree maps provide a visually engaging and informative way to uncover valuable insights.

4.4. Waterfall Charts

Waterfall charts are an advanced visualization tool used to illustrate the cumulative effect of sequentially added or subtracted values on a starting point. They are especially helpful in breaking down a total into its contributing components, making them widely used in financial analysis, operational reporting, and performance tracking. By showing how individual elements influence an overall outcome, waterfall charts provide a clear and comprehensive understanding of how a value progresses from start to finish.

One of the key strengths of a waterfall chart is its ability to visualize both positive and negative changes, offering a balanced perspective on performance. For example, in financial scenarios, a waterfall chart can show how costs, discounts, and taxes reduce revenue, while additional income streams or gains add to it, eventually leading to the net profit. This makes it an invaluable tool for presenting complex data in a structured and easy-to-digest format.

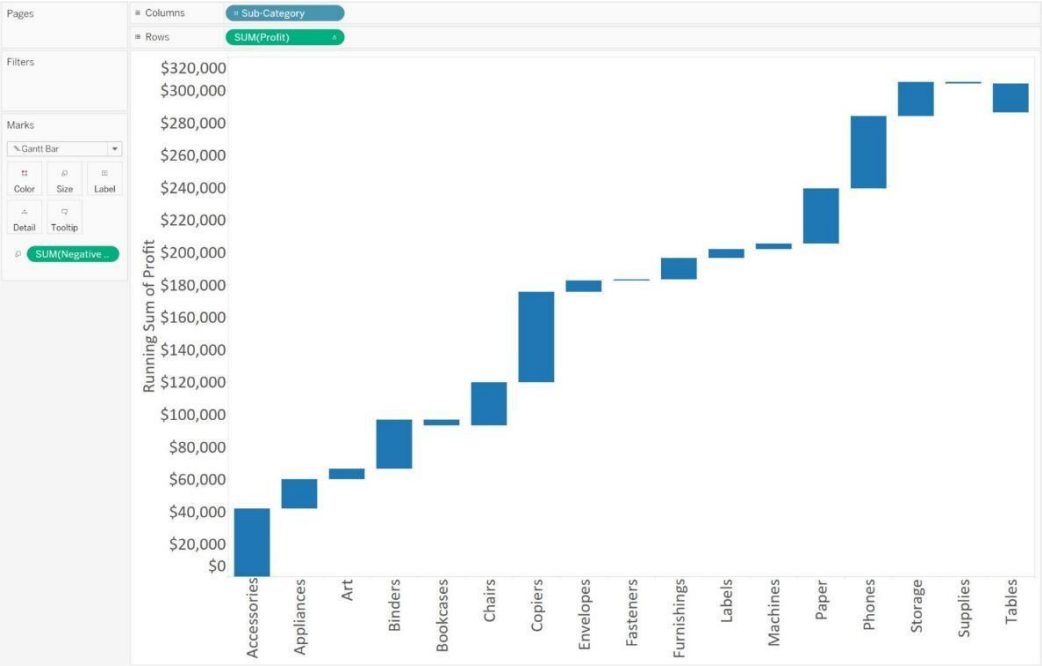


Fig 8.6 *Example of a Waterfall Chart*

Steps to Create a Waterfall Chart in Tableau:

Follow these steps to create a waterfall chart in Tableau:

1. **Set the Foundation with Data:**

   Drag the measure representing the cumulative value (e.g., Profit) to the Rows Shelf. This measure will reflect the changes in values over different categories or factors.

2. **Add the Dimension for Categories:**

   Drag a dimension (e.g., Category) to the Columns Shelf. This divides the cumulative effect into specific contributing categories, creating individual segments for each.

3. **Apply a Running Total Calculation:**

   Click on the Profit pill in the Rows Shelf and select Quick Table Calculation > Running Total. This transforms the measure into a cumulative value, showcasing how the total changes as each category is added or subtracted.

4. **Adjust the Marks Card for Differentiation:**

   - Change the Marks type to Gantt Bar to represent the incremental changes.
   - Drag the same measure (Profit) to the Size card to adjust the size of the bars.
   - Add a calculated field to differentiate between positive and negative contributions, such as: IF [Profit] > 0 THEN "Positive" ELSE "Negative" END

- Drag this calculated field to the Color card to visually distinguish increases (e.g., green) from decreases (e.g., red).

5. **Format the Chart:**
   - Customize labels, colors, and formatting to improve readability.
   - Add reference lines or annotations if needed for additional insights.

Example: Profit Analysis for a Retail Company

A retail company wants to analyze its profit over several categories to understand the contribution of costs, discounts, and other factors.

1. Data Setup:
   - Measure: Profit
   - Dimension: Category (e.g., Furniture, Technology, Office Supplies)
2. Chart Construction:
   - The waterfall chart starts with total revenue as the base value.
   - Categories like "Discounts" and "Shipping Costs" contribute negative values, reducing the overall profit.
   - Categories such as "Technology Sales" and "Office Supplies" contribute positive values, increasing the profit.
3. Insights:

   The chart visually shows how each category impacts the profit. For instance, high shipping costs might significantly reduce revenue, while technology sales might compensate with a large positive impact.

   The final bar represents the net profit after all contributing factors have been accounted for.

Waterfall charts are an indispensable tool for visualizing cumulative effects across categories. By breaking down total values into their constituent parts, they help businesses understand the drivers of performance and identify areas of improvement. Whether analyzing profits, costs, or any sequential process, waterfall charts provide a structured and intuitive way to communicate complex data insights.

4.5. Sankey Diagrams

Sankey diagrams are specialized flow diagrams that visually represent the movement or flow of quantities between different nodes or categories. The width of the connecting paths or flows is

proportional to the quantity they represent, making it easy to see where resources, values, or processes are distributed. These diagrams are useful for visualizing systems where data flows from one stage to another, such as supply chains, financial transfers, or website traffic analysis.

The key advantage of Sankey diagrams lies in their ability to display complex relationships and distributions intuitively and aesthetically pleasingly. For example, in energy management, a Sankey diagram can illustrate how energy is generated, consumed, and lost at various stages of production and distribution. Similarly, businesses can use these diagrams to track customer journeys, showing how users move through different touchpoints before converting. Their ability to present intricate connections at a glance makes them invaluable for exploratory data analysis and storytelling.

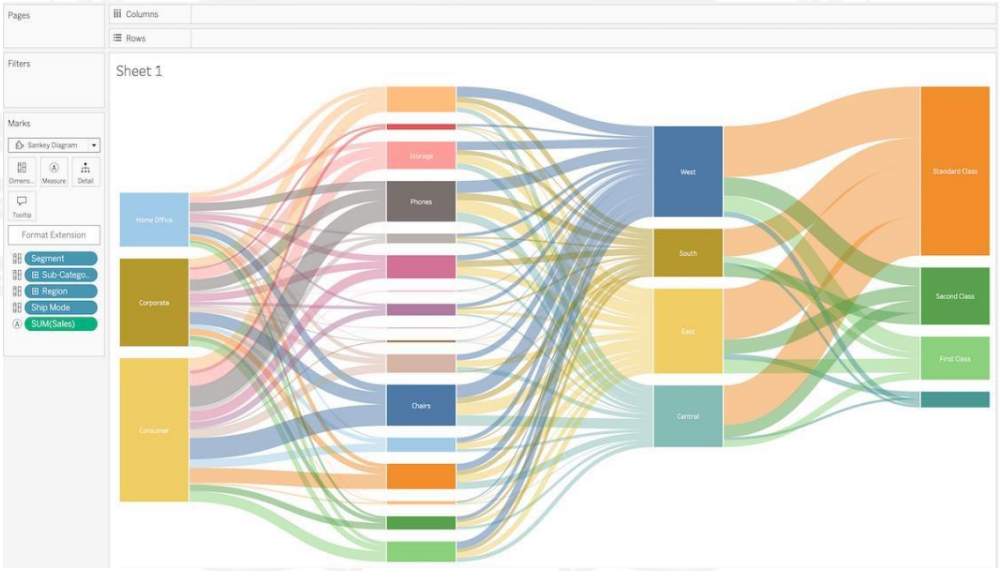


Fig 8.7 *Example of a Sankey Diagram*

Steps to Create a Sankey Diagram in Tableau:

While Tableau does not have a native Sankey diagram feature, you can create one using calculated fields, data preparation, and creative use of path-based visualizations. Below are the detailed steps:

1. **Prepare the Data:**

   The dataset should include:

   - Source: The starting point of the flow (e.g., Region, Department).
   - Target: The endpoint of the flow (e.g., Category, Sub-category).
   - Value: The quantity to represent (e.g., Sales, Volume).

   Add a "Path" field to simulate the flow curves. Typically, a spreadsheet tool or scripting will generate intermediate points for smooth connections.

2. **Load Data into Tableau:**

   Import the prepared dataset into Tableau. Ensure all necessary fields, including Source, Target, Value, and Path, are correctly formatted.

3. **Create a Calculated Field for Pathing:**

   Use a calculated field to define the curvature of the flows. For example:

   SIN(PI() * [Path] / MAX([Path]))

   This formula helps create a smooth, curved transition between nodes.

4. Build the Visualization:
   - Drag Path to the Columns Shelf.
   - Drag the calculated field for flow (e.g., Curvature) to the Rows Shelf.
   - Use Source and Target as dimensions on the Detail card to define the start and end nodes.
   - Place a Value on the Size card to determine the thickness of the flows.
5. Customize the Diagram:
   - Assign distinct colors to Source and Target categories using the Color card.
   - Format the chart to enhance readability by adjusting axis ranges and adding labels to nodes and flows.

Example: Website Traffic Flow Analysis

A digital marketing team wants to understand user behaviour on their e-commerce website, tracking how visitors navigate from landing pages to final conversions.

Data Setup:

- Dimensions: Source (e.g., Landing Page, Product Page, Cart) and Target (e.g., Product Page, Cart, Checkout).
- Measure: Visitors (number of users transitioning between stages).

Sankey Diagram Visualization:

- The Sankey diagram starts with a wide flow from the "Landing Page," representing the total number of visitors.
- Smaller flows split from the landing page to "Product Page" and "Cart," reflecting user actions.
- The narrowest flow reaches "Checkout," showing the final conversion rate.
- Different colors for each stage help distinguish between user interactions.

Insights:

The team observes that many users drop off after the "Product Page," indicating potential issues with the page's content or navigation.

They also see that users who proceed to the "Cart" are more likely to convert, suggesting an opportunity to optimize the checkout process further.

Sankey diagrams are a powerful visualization tool for understanding and communicating flow-based relationships. By effectively representing the magnitude and direction of connections between nodes, they provide valuable insights into systems where movement or distribution is key. Whether analyzing energy use, financial transactions, or customer journeys, Sankey diagrams enable organizations to uncover patterns, optimize processes, and make data-driven decisions.

4.6. Scatter Plot with Trend Lines

Scatter plots are a fundamental visualization technique that displays relationships between two continuous variables. By plotting individual data points on an X-Y axis, scatter plots reveal patterns, clusters, or outliers that might otherwise remain hidden in raw data. The addition of trend lines enhances this analysis by summarizing the overall relationship between the two variables, making it easier to discern correlations or trends at a glance.

Trend lines, often calculated using regression techniques, serve as visual guides that connect the dots—literally and figuratively. They provide insights into the strength and direction of the relationship between variables. For instance, a positive trend line in a scatter plot of sales and profit suggests that higher sales are generally associated with higher profits. Meanwhile, identifying outliers in the scatter plot—points that deviate significantly from the trend—can lead to further analysis and discovery of unique or problematic cases.

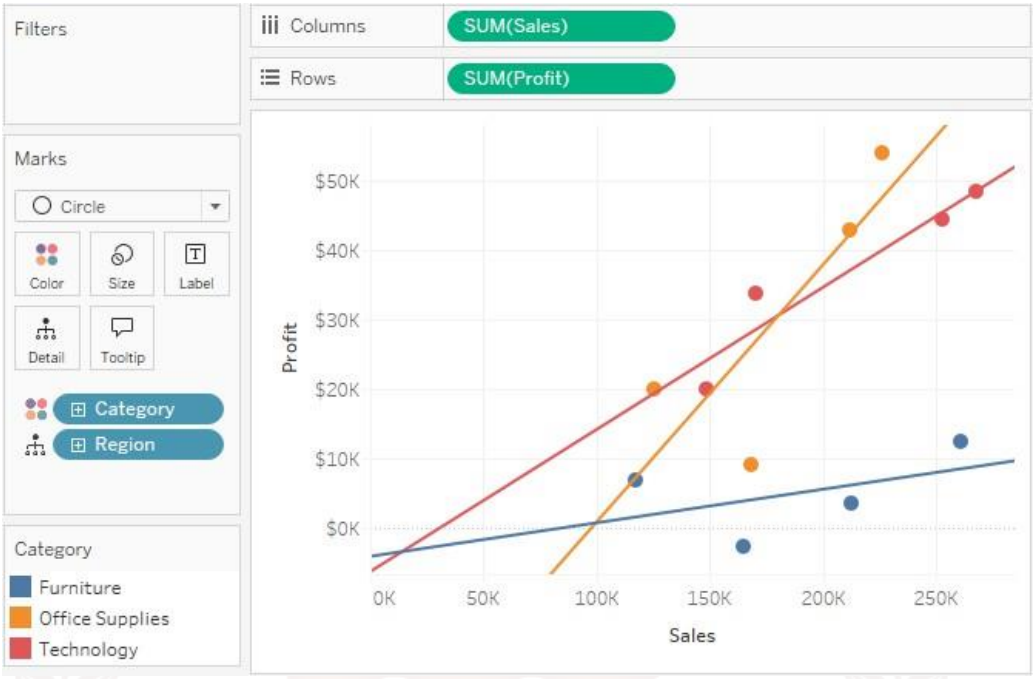


Fig 8.8 *Example of a Scatter Plot with trendlines*

Steps to Create a Scatter Plot with Trend Lines:

1. Set Up the Measures:
   - Drag a continuous measure, such as Sales, to the Columns Shelf. This forms the X-axis of the scatter plot.
   - Drag another continuous measure, such as Profit, to the Rows Shelf, creating the Y-axis.
2. Add Dimensions for Granularity:

   Drag a dimension, such as Region or Product Category, to the Detail Card on the Marks Shelf. This differentiates data points based on the chosen dimension, enabling more granular insights.
3. Enable Trend Lines:
   - Open the Analytics Pane and select Trend Line. Drag it onto the scatter plot.
   - Choose the type of trend line (e.g., linear, exponential, logarithmic) based on the nature of the data and the relationship being analyzed.
4. Customize the Visualization:
   - Use the Color Card to color-code points based on categories or regions for enhanced interpretability.
   - Add labels to display specific data values for outliers or clusters of interest.
   - Format the axes, gridlines, and titles to improve readability.

Example: Analyzing Sales and Profit Correlation by Product Category

A retail company wants to explore the relationship between sales and profit across different product categories to identify areas of strength and weakness.

Data Setup:

- Measures: Sales (X-axis) and Profit (Y-axis).
- Dimension: Category.

Scatter Plot Visualization:

- Each point in the scatter plot represents a product category, plotted based on its sales and profit values.
- The trend line, added using the Analytics pane, shows a positive slope, indicating that higher sales are generally associated with higher profits.

Insights:

- Most categories cluster along the trend line, confirming the expected relationship between sales and profit.
- An outlier is identified: a category with high sales but relatively low profit. This suggests higher costs or discounts for products in this category.
- Another outlier, with low sales and high profit, indicates a niche product with high margins, which could be an area for strategic expansion.

Scatter plots with trend lines are invaluable for exploring relationships between two continuous variables. The trend line simplifies complex datasets into actionable insights by highlighting correlations and deviations. Whether used for financial analysis, operational performance, or customer behavior, scatter plots and trend lines enable data-driven decision-making by providing a clear visual representation of patterns and anomalies.

4.7. Combining Techniques

The true power of Tableau lies in its ability to combine various features and techniques to create interactive, insightful, and visually compelling dashboards. By blending calculated fields, parameters, and advanced charts, analysts can unlock the full potential of their data. These combinations allow for more nuanced exploration, enable user-driven customization, and enhance storytelling capabilities. Instead of isolated visualizations, users can design dynamic dashboards that adapt to specific business questions or scenarios, catering to diverse audiences.

For instance, a single visualization can evolve into an interactive tool that empowers decision-makers. When paired with a parameter, a bullet chart can allow users to switch benchmarks dynamically, helping them evaluate performance across different criteria. Similarly, integrating a heat map with calculated fields can automatically highlight regions based on performance thresholds, making it easier to identify areas of focus. This synergy between Tableau's features streamlines analysis and delivers actionable insights.

Combining Techniques: Steps and Applications

1. **Calculated Fields for Custom Metrics:**
   - Calculated fields act as the foundation for advanced analytics. They allow users to create custom metrics tailored to specific business needs, such as profit margins, growth rates, or performance indices.
   - Example: Create a calculated field called Performance Score using a formula like:
     IF [Sales] > 50000 THEN "High"
     ELSEIF [Sales] > 20000 THEN "Medium"
     ELSE "Low"
     END
2. **Parameters for Interactivity:**
   - Parameters let users interact with visualizations by modifying values in real-time. These user inputs can be tied to calculated fields or used to control filters, reference lines, or display settings.
   - Example: Add a parameter to let users select benchmarks (e.g., 50,000, 75,000, 100,000) for a bullet chart, allowing dynamic comparisons.
3. **Advanced Charts for Impactful Storytelling:**
   Advanced charts, such as bullet charts, heat maps, or scatter plots, serve as powerful storytelling tools. When combined with interactivity and calculated fields, they become even more engaging and informative.
4. **Integration:**
   Combine these techniques for a cohesive, interactive dashboard:
   - Use a bullet chart to show actual sales versus target values.
   - Add a parameter to enable users to switch between different target benchmarks.

- Include a heat map that highlights regions exceeding their sales goals using a calculated field, like:

  IF [Sales] >= [Benchmark Parameter] THEN "Above Target"

  ELSE "Below Target"

  END

Example: Combining Techniques to Evaluate Regional Performance

A retail company wants to analyze regional performance, comparing sales against dynamically adjustable benchmarks and highlighting regions that meet or exceed targets.

Setup:

- Calculated Field: Create a field called Performance Status to classify regions as “Above Target” or “Below Target” based on sales:

  IF [Sales] >= [Benchmark Parameter] THEN "Above Target"

  ELSE "Below Target"

  END
- Parameter: Create a Benchmark Parameter with 50,000, 75,000, and 100,000 options.
- Bullet Chart: Plot sales for each region with a reference line controlled by the Benchmark Parameter.
- Heat Map: Visualize performance across regions, using the Performance Status calculated field to determine color intensity.

Visualization:

- A bullet chart dynamically adjusts the benchmark line based on the user-selected parameter, allowing users to see which regions exceed or fall short of the target.
- A heat map highlights regions with color-coded intensity: green for “Above Target” and red for “Below Target.”
- Labels and filters let users focus on specific regions or adjust benchmarks for tailored analysis.

Users can create powerful, dynamic visualisations beyond static reporting by combining Tableau techniques such as calculated fields, parameters, and advanced charts. These integrations allow for deeper insights and more engaging storytelling, enabling stakeholders to interact with the data and uncover actionable trends. Whether analyzing performance, identifying outliers, or driving business decisions, the combination of techniques in Tableau transforms raw data into an interactive narrative.

5. SUMMARY

The chapter on data visualization explores the significance, applications, and types of this powerful technique for representing information visually. Data visualization is pivotal in transforming complex data sets into comprehensible visuals, aiding in quicker insights and informed decision-making.

The applications of data visualization are diverse and far-reaching. It helps doctors comprehend patient histories rapidly, leading to improved emergency care. In business intelligence, cloud-connected analytics facilitate data-driven decision-making. Military applications involve rapidly delivering accurate information for informed action, while finance industries benefit from visualizing customer behavior and investment strategies.

The chapter delves into the different types of data visualizations. Comparison visualizations, such as bar charts and population pyramids, showcase variations between values. Pattern visualizations, like line graphs and scatter plots, reveal trends and relationships in data. Changes visualizations, like candlestick charts, focus on displaying changes over time, particularly in financial analysis.

Lastly, the chapter discusses conceptual visualizations, such as flowcharts and mind maps, that represent abstract concepts and processes. It explores planning and calendar visualizations, including calendars and timelines, which provide insights into scheduling and temporal relationships.

In summary, the chapter underscores the pivotal role of data visualization in transforming complex data into actionable insights across various domains. It highlights its applications and showcases the diversity of visualization types contributing to better understanding, informed decision-making, and effective communication of data-driven information.

6. GLOSSARY

Data Visualization	-	It is the graphical representation of data and information using visual elements like charts, graphs, and maps. It helps simplify complex data for better understanding and decision-making.
Business Intelligence (BI)	-	A technology-driven process for analyzing data and presenting actionable insights to help executives, managers, and other corporate end users make informed business decisions.
Categorical Data	-	Data can be divided into distinct groups or categories, such as gender, location, or product type. Bar charts or pie charts often represent it.
Ordinal Data	-	Data representing categories with a meaningful order or ranking but without a specific numerical value, such as customer satisfaction ratings (e.g., 1-5 stars). It can be visualized with ordered charts or bar graphs.
Numerical Data	-	Quantitative data that can be measured and expressed numerically, such as sales figures, temperatures, or ages. This data type is often visualized using histograms, line graphs, or scatter plots.
Temporal Data	-	Data that relates to time, such as timestamps, dates, or durations. This data type is typically displayed using time series, line, or area charts.
Interactive Dashboards	-	A visual interface that displays data from multiple views allows users to interact with the data, filter information, and drill down into specific insights.
Citizen Data Scientist	-	A non-expert in data science who uses tools and technologies to analyze data, create models, and extract insights. These individuals typically work in business roles rather than technical data science roles.
Data Storytelling	-	It is the practice of using data visualizations and narratives to convey a story that communicates insights and engages the audience. It helps make data more relatable and understandable.
Highlighting	-	The process of emphasizing specific data points or trends in a dataset using visual cues such as color or size to help the viewer identify important patterns or anomalies.

Filter Action	-	A feature in data visualization tools that allows users to filter data across multiple views or dashboards based on their selections, enabling more focused data exploration.
Go to Sheet Action	-	This is a type of action in dashboards that allows users to navigate from one sheet or view to another, enhancing interactivity and providing access to more detailed data.
URL Action	-	A feature that lets users click on a mark or link in a dashboard to open a web page, providing external context or additional information related to the selected data.
Heat Map	-	A data visualization that uses color to represent values in a matrix, heat maps are often used to show the intensity of data points across a geographic area or within a dataset.
Scatter Plot	-	A type of data visualization that uses points on a graph to represent the values of two variables, helping to identify relationships or correlations between them.
Dashboard Actions	-	Interactive elements in dashboards that allow users to filter, highlight, or navigate through different views, enhancing the overall user experience and enabling more dynamic exploration of data.
Real-time Data Visualization	-	The real-time display of data allows users to monitor current trends or performance metrics as they change.
Bar Chart	-	A graphical representation of data where rectangular bars represent the frequency or value of a variable, often used for categorical data.
Pie Chart	-	A circular chart divided into slices to illustrate numerical proportions, commonly used to represent parts of a whole.
Line Chart	-	A graph that displays data points connected by straight lines, often used for showing trends over time in temporal data.

7. SELF-ASSESSMENT QUESTIONS

Fill in the Blanks:	
1	Scatter plots show relationships between _____ variables.
2	Heat maps use _____ gradients to represent intensity.
3	Calculated fields help in transforming _____ data into insights.
True or False	
4	Parameters can hold multiple values at once.
5	Waterfall charts only show positive values.
6	Sankey diagrams represent flow and movement of data.
7	Bullet charts are useful for performance analysis.
Multiple Choice Questions:	
8	Which of the following is a key advantage of using calculated fields in Tableau? A) Reducing dataset size B) Creating new insights from existing data C) Automating data collection D) Automating data collection
9	Which Tableau feature is best suited for performing “what-if” analysis? A) Filters B) Parameters C) Worksheets D) Trend lines
10	In a Sankey diagram, what does the width of the flow represent? E) Category type F) Data labels G) Magnitude of flow or quantity H) Time duration

8. TERMINAL QUESTIONS

Short Answer Questions

1. Differentiate between calculated fields and parameters in Tableau with one example each.
2. List any four types of calculations in Tableau and briefly explain their use.
3. How do colour and size encoding enhance data interpretation in heat maps?
4. State any four advantages of using advanced visualisations over basic charts in Tableau.
5. What are the key components required to create a tree map in Tableau?

Long Answer Questions

1. Analyse the role of parameters in enhancing user interactivity and dashboard flexibility in Tableau.
2. Explain how calculated fields improve data transformation and decision-making in Tableau dashboards.
3. Analyse the application of waterfall charts in financial and business performance analysis.
4. Discuss the advantages and limitations of advanced visualisation techniques in Tableau.

9. ANSWERS

Self-Assessment Questions

1. two
2. colour
3. raw
4. False
5. False
6. True
7. True
8. B) Creating new insights from existing data
9. B) Parameters
10. C) Magnitude of flow or quantity

Terminal Questions Answers

Short Question Answers

1. Calculated fields are used to create new data from existing data using formulas and logic. They allow users to perform operations such as arithmetic calculations, conditional logic, and aggregations. For example, profit margin can be calculated using: Profit Margin = Profit / Sales.

   This helps in generating new insights directly from the dataset. Parameters, on the other hand, are user-controlled input values that allow dynamic interaction with dashboards. They are single-value variables that can influence calculations and visualisations. For example, a parameter can allow users to switch between viewing Sales, Profit, or Discount in a chart. The key difference is that calculated fields are data-driven, while parameters are user-driven. Calculated fields process existing data, whereas parameters modify how the data is displayed or analysed.

2. **Tableau supports various types of calculations that help in analysing and transforming data.**
   1. Arithmetic Calculations – Perform basic operations like addition, subtraction, multiplication, etc.
      Example: Sales – Discount.
   2. Logical Calculations – Use conditions such as IF and CASE statements to categorise data.
      Example: IF Sales > 50000 THEN “High” ELSE “Low”.
   3. String Calculations – Used to manipulate text data, such as combining first and last names.

4. Date Calculations – Perform operations on date fields.

   Example: DATEDIFF to calculate delivery time.

3. Colour and size encoding play a significant role in enhancing data interpretation in heat maps. Colour encoding represents the intensity or magnitude of data values. Darker or warmer colours indicate higher values, while lighter or cooler colours represent lower values. This helps users quickly identify patterns, trends, and outliers. Size encoding adds another layer of information by varying the size of elements based on a measure. Larger sizes indicate higher values, while smaller sizes represent lower values. Together, colour and size improve visual clarity and allow users to interpret complex datasets efficiently. They make heat maps more informative and support faster decision-making.
4. **Advanced visualisations offer several advantages over basic charts:**
   A) Better Insight Generation – They reveal complex patterns and relationships that simple charts cannot.
   B) Improved Data Storytelling – Advanced charts communicate insights more effectively.
   C) Handling Large Data – They can represent large and complex datasets clearly.
   D) Enhanced Interactivity – Features like parameters and filters make dashboards dynamic. Overall, advanced visualisations improve analysis, engagement, and decision-making.

5. A tree map requires the following key components:

A) **Primary Dimension** – Represents the main category (e.g., product category).
B) **Sub-Dimension** – Provides hierarchical detail (e.g., sub-category).
C) **Measure for Size** – Determines the size of rectangles (e.g., sales).
D) **Measure for Colour** – Represents another metric (e.g., profit).

Long Question Answers

1. Parameters in Tableau play a crucial role in enhancing user interactivity and dashboard flexibility by allowing users to dynamically control data visualisations. Unlike filters, parameters are not limited to restricting data; instead, they enable users to modify calculations, switch between metrics, and simulate scenarios. One of the key contributions of parameters is dynamic interaction. Users can input values through sliders, dropdowns, or text boxes, which immediately update the visualisation. For example, a parameter can allow users to switch between viewing sales, profit, or discount within a single chart. This reduces the need for multiple visualisations and makes dashboards more efficient. Parameters also enhance flexibility by enabling "what-if" analysis. Users can adjust variables such as budget, target sales, or discount rates to see how outcomes change. This is particularly useful in strategic decision-making, as it allows organisations to test different scenarios without altering the underlying data.

   Another important role is their integration with calculated fields. Parameters can be combined with calculations to create dynamic logic. For instance, a parameter can define thresholds for highlighting high-performing regions or dynamically change the metric displayed.From a user experience perspective, parameters improve engagement and usability. They make dashboards interactive, allowing users to explore data based on their specific needs. This personalised interaction enhances understanding and encourages deeper analysis. However, parameters must be used carefully. Too many parameters can overwhelm users and reduce dashboard clarity. Therefore, it is important to limit options and use clear labels.

2. Calculated fields are a powerful feature in Tableau that enable users to transform raw data into meaningful insights, thereby improving decision-making. They allow the creation of new metrics using formulas, logical expressions, and aggregations. One of the primary roles of calculated fields is data transformation.

   They enable users to manipulate data by combining, filtering, or converting existing fields. For example, calculating profit margin (Profit / Sales) provides deeper insight into business performance beyond basic sales figures. Calculated fields also support advanced aggregation. Users can compute averages, running totals, and ratios, which help identify trends and patterns.

   This enhances analytical depth and provides a clearer understanding of data. Another important function is conditional logic. Using IF or CASE statements, calculated fields can categorise data into meaningful groups, such as "High Value" and "Low Value" customers. This segmentation

supports targeted strategies and better decision-making. Calculated fields improve visualisation customisation by enabling dynamic formatting, labelling, and highlighting. For instance, underperforming regions can be highlighted automatically, helping managers focus on critical areas. From a decision-making perspective, calculated fields provide actionable insights. They convert raw data into meaningful metrics that support strategic planning, performance evaluation, and forecasting.

3. Waterfall charts are widely used in financial and business analysis to illustrate how an initial value changes through a series of positive and negative contributions. They provide a clear visual representation of cumulative effects. In financial analysis, waterfall charts are particularly useful for profit and loss analysis. They show how revenue is affected by factors such as costs, discounts, taxes, and other expenses. For example, a company can analyse how sales revenue is reduced by operational costs and increased by additional income streams, ultimately leading to net profit. Another application is in budget analysis and forecasting.

   Waterfall charts help organisations understand how different factors impact budgets. By visualising incremental changes, managers can identify areas of overspending or cost savings. In business performance analysis, waterfall charts are used to evaluate key performance drivers. They help identify which factors contribute positively or negatively to performance. For instance, in sales analysis, factors such as promotions, returns, and logistics costs can be visualised clearly. One of the key advantages of waterfall charts is their ability to provide clarity and transparency. They break down complex data into understandable components, making it easier for stakeholders to interpret results.

   They also support decision-making by highlighting critical areas that require attention. For example, high operational costs can be identified and addressed. Waterfall charts may become complex when too many categories are included, reducing readability. Proper formatting and simplification are essential. These charts are powerful tools for financial and business analysis. They provide clear insights into cumulative changes, improve understanding of performance drivers, and support strategic decision-making.

4. Advanced visualisation techniques in Tableau offer significant advantages in analysing complex datasets and improving data storytelling. However, they also have certain limitations that must be considered. One of the major advantages is enhanced insight generation. Advanced charts such as heat maps, tree maps, and Sankey diagrams reveal patterns, relationships, and trends that are not easily visible in basic charts. Another advantage is improved data storytelling.

   These visualisations present data in a visually appealing and intuitive manner, making it easier for users to understand insights. For example, heat maps quickly highlight high and low performance areas using colour gradients. Advanced visualisations also support handling large and complex datasets. They allow users to analyse multiple variables simultaneously, providing a comprehensive view of data. Additionally, they enable interactivity and dynamic analysis when combined with features like parameters and calculated fields. This enhances user engagement and allows customised exploration.

   There are some limitations. Advanced visualisations can be complex to design and interpret, especially for non-technical users. They may require additional training and expertise. Another limitation is performance issues. Complex calculations and large datasets can slow down dashboards, affecting usability.

   There is also a risk of overloading visualisations, where too many elements reduce clarity and make interpretation difficult. To conclude, advanced visualisation techniques provide powerful tools for data analysis and storytelling. While they offer significant advantages in insight generation and interactivity, careful design and optimisation are necessary to overcome their limitations and ensure effective communication.

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